Coding & Solution

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	GitHub repository link to be added after final submission.
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	• Crop recommendation using Machine Learning • Real-time crop prediction • Soil nutrient and weather data analysis • User-friendly Flask web interface • Responsive design
Pending / Incomplete Features	Cloud deployment and weather API integration.
Setup / Run Instructions	Install the required packages from requirements.txt, train the model (if needed), and run app.py to start the Flask application.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions/modules.	Yes
2	Meaningful variable and function names are used.	Yes
3	Code includes comments where necessary.	Yes
4	Error handling is implemented for user input.	Yes
5	Application runs without critical errors.	Yes
6	Code is committed to a version control repository(Github).	Yes

Additional Notes/Comments
The project follows a modular Flask architecture with separate folders for templates, static files, dataset, and the trained Machine Learning model. The implementation emphasizes readability, maintainability, and reusability, making it easier to update and extend in the future.